An intelligent Surveillance System for Detecting Abnormal Behaviors on Campus using YOLO and CNN-LSTM Networks

Abdallah Khayrat[1], Pierre Malak[1], Maryam Victor[1], Seifeldin Ahmed[1], Haytham Metawie[1], Verina Saber[1], and Mohammad Elshalakani[2]

[1]Faculty of Computer Science, Misr International University, Cairo, Egypt
[2]Faculty of Engineering, Ain Shams University, Cairo, Egypt
[1]abdallah181280, pierre1805213, maryam1805897, seifeldin1802862, haytham.metawie, verina.saber@miuegypt.edu.eg
[2]momar@eng.asu.edu.eg

2022 2nd International Mobile, Intelligent, and Ubiquitous Computing Conference (MIUCC) | 978-1-6654-6677-6/22/$31.00 ©2022 IEEE | DOI: 10.1109/MIUCC55081.2022.9781786

***Abstract*—Like many other surveillance systems, security and safety are mainly what concern people these days. It can be used in two methods, watching threats in real-time or searching for an event that already happened. Surveillance systems can be defined in many different ways like violence or any abnormal behaviour detection. In this paper, we propose a surveillance system for detecting abnormal behaviours on campus using deep learning models. It also includes how YOLO, CNN, and LSTM are used to apply concepts like feature extraction, object detection, action detection and identification. YOLO algorithm was used in the smoking detection and the playing cards achieving the accuracy of 90% and 93.8% respectively. As for the CNN-LSTM model, we achieved an accuracy of 93.5%.**

***Index Terms*—Surveillance, YOLO, CNN, LSTM, Deep Learning, Object Detection, Fight Detection**

I. Introduction

Detecting human behaviours gained a lot of attention recently in the field of artificial intelligence and deep learning fields. Deep learning models cement detecting specific activities by learning from photos or videos that contain the action we are looking for, as well as its contradiction.[1] Activity/behaviour detection can be used in a wide range of environments. A whole lot of applications depend on surveillance and deep learning such as Amazons's go store[2], theft detection, violence detection and any activity that needs detection to trigger something accordingly.

Every university strives to keep its campus environment safe from deviant and unethical behaviour. Detecting such violations and recognising rules-breakers among students, is a difficult process in universities, especially when relying solely on human resources. That inspired us to propose our system, whose substantial idea is to utilise learning algorithms to detect unethical behaviour among students on campus. And by extension identifying the rule-breaker, and issuing tickets to him/her.

978-1-6654-6677-6/22/$31.00 ©2022 IEEE

We used YOLOv5[3] algorithm to detect smoking and playing cards. The object detection method YOLO, which stands for "You Only Look Once,"[4] separates images into a grid structure. Each grid cell is in charge of detecting items within itself. For its speed and precision, YOLO is one of the most well-known object recognition algorithms.

Long short-term memory (LSTM) [5][6] is a deep learning architecture based on an artificial recurrent neural network (RNN). LSTM has feedback connections, unlike normal feedforward neural networks. It can handle not only individual data points (such as photos) but also complete data streams (such as speech or video). LSTM can be used for tasks like unsegmented, linked handwriting recognition, speech recognition, and anomaly detection in network traffic or IDSs, for example (intrusion detection systems). A cell, an input gate, an output gate, and a forget gate to form a common LSTM unit. The three gates control the flow of information into and out of the cell, and the cell remembers values across arbitrary time intervals. The algorithm is used to detect one of the main unethical and unwanted behaviour such as fighting detection.

In this paper, our contribution is represented as follows:

- Providing an intelligent surveillance system for detecting abnormal behaviours on-campus, its intelligence comes from using deep learning models in detecting unethical activities between university students on campus.
- We utilized Convolutional Neural Networks (CNN)[1] and Long short-term Memory (LSTM) for fighting detection between students.
- Using Yolov5 Architecture for smoking detection and playing cards detection.

The rest of the paper is organized as follows: Section II covers similar systems by explaining how they work and what they used to achieve their work. In section III, we vividly explain our workflow and the methods we have chosen to work with as well as the data we use. Section IV shows the experimental results we obtained from the models that were

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 04,2026 at 03:57:59 UTC from IEEE Xplore. Restrictions apply.

developed, and at last we conclude briefly.

II. Related work

This section describes the research study for the same domain. We divided up and grouped the related work into four categories based on the detection interest. The first category represents the work done in facial recognition, The second one covers fight detection, the third represents smoking detection and at last the fourth discusses playing card detection.

A. Face and speech recognition

In the metacarpus Personal Assistant Robot, Speech detection was provided by using the Google Speech Recognition API to transform speech to text and then match it with commands if they are relevant. For facial detection, Open-cv is utilised; their robot is first educated with the owner's face data, after which the data is sent to a central computer with a deep learning face recognition model with an accuracy of 99.38% [7].

The authors here [8] intended to recognise all of the people in a video frame they used the OpenCV Haar-Cascade method to implement the cascade of classifiers concept. They used a set of photos to train their system to detect them.

The aim of this study is to develop a reliable face recognition system that decreases computing costs while increasing accuracy by incorporating iris recognition with face recognition. OpenCV and Python tools are used to manipulate iris and facial images. Such algorithm compares all of the histograms and generates the optimal label and confidence, resulting in better face recognition. For identification, iris technologies capture the unique feature of the iris in the human eye. The LBHBR face models are compared using the Haar-Cascade classifier.[9]

B. Fighting detection

For surveillance systems, Ersin ESEN, Mehmet Ali ARABACI, and Medeni SOYSAL designed a Fight detection algorithm. They suggested an approach based on the Motion Co-Occurrence Feature, which is a novel motion feature (MCF). The proposed technique has been tested using a video collection that includes samples from the widely used and publicly available BEHAVE[10] and CAVIAR[11] databases, as well as popular videos found on the Internet. The detection performance is computed using 10-fold cross-validation with the kNN classifier, and they report a 93.5% accuracy.[12]

Sarah Almeida Carneiro, Gabriel Pellegrino da Silva, Silvio Jamil F. Guimaraes and Helio Pedrini aim to perform detection of actual fights on surveillance recordings. They propose the design of a CNN VGG16. High-level Feature extraction occurs first then the weight calculation using CNN then they use SVM to merge the results of each stream for classification. The details of their proposed approach are 10-fold cross-validation, a batch size of 1024, a threshold of 0.5 and 500 epochs. The paper introduces an effective architecture for a fight detection model and details the model's hyperparameters [13].

This research developed a deep representation based model for detecting violent scenes utilising the concept of transfer learning to recognise aggressive human behaviour. The GoogleNet model has fewer parameters than the AlexNet model, which they employ as a pre-trained model. As a 2D deep CNN network, it learns spatial information from images. It is fine-tuned using transfer learning on the Hockey and Movies datasets to develop a deep representation classifier for identifying violent scenes. The tests are carried out on two benchmark datasets for detecting violent activity: Hockey and Movies[14]. Firstly they uses the Hockey dataset, which contains 1000 360x288-resolution video clip of violent entanglements among Hockey players. And due to a broad array of scenarios, they used the Movies dataset as it contains many complications, including variations in the background with different lighting conditions and occlusions.[15]

C. Smoking detection

Online videos have become increasingly popular as Internet technology has advanced and network quality has improved. In recent years, online live broadcast has become a hotspot, and smoking behaviour in these broadcasts is damaging to both smokers and the environment. As a result, detecting and efficiently controlling smoking behaviours in video content is required. Traditionally, They develop a smoking detection system that introduces a smoking picture recognition model that uses a convolutional neural network to detect smoking behaviours in video footage using images. Their idea was to do this because Traditionally, smoking images are detected based on the detection algorithms of cigarette smoke. Given the limited resolution of live broadcast videos, cigarette smoke is not visually apparent in the video content. This approach should detect smoking images without requiring the detection of cigarette smoke by relying solely on information from human smoking movements and cigarette image characteristics. They compared the results of different pre-trained models and the best precision, recall, accuracy and f1 score they got was 90% each from the Smoking Net model. This paper is so useful because it suggests the most optimum approach and a more convincing and realistic way of detecting cigarette smoking action. It also compares a lot of popular models with all the important metrics.[16]

Cigarette smoke degrades air quality, which harms human health. So they thought it was critical to have a system that can detect the presence of dangerous gases in cigarette smoke to produce a smoke-free workplace. The hardware they use is an ESP8266 ESP-01 Wi-Fi module and four gas sensors, mainly MQ-135, MQ-2, MQ-7, and MQ-9, to sense, collect, and transmit data wirelessly to the Raspberry Pi. The software they develop is for data collection, logging, and retrieval. They build Python software that runs on the Raspberry Pi

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 04,2026 at 03:57:59 UTC from IEEE Xplore. Restrictions apply.

for data logging. The application uses the MQTT protocol to accept data from a sensor node and sends it to a webserver over the Internet. For data retrieval, the webserver utilizes the Hypertext Preprocessor (PHP). They state that the user can use a web browser to access sensor data from the webserver. The communication protocol between a sender (NodeMCU ESP8266) and a receiver (Raspberry Pi) is Message Queuing Telemetry Transport (MQTT), a publish-subscribe-based messaging system. To handle wires, a separate software that implements the MQTT protocol is uploaded to NodeMCU.[17]

Since sites with a high need for safety, such as gas stations, have a high demand for surveillance of harmful behaviours like smoking or using cell phones. This study proposes a deep learning solution to replace the former method, in which gas stations would employ appropriate staff to monitor and supervise harmful behaviours. They believe that it is neither accurate nor cost-effective. To address the aforementioned issues, this article explains how to develop a deep learning-based object detection system. The target object is detected in this study by training a convolutional neural network (CNN). The research uses a Linux-based development board to capture photos or images captured using an external UVC camera, and then processes them for detection. They claim that when compared to other neural networks, this study performs well in terms of accuracy and speed. The drawback of this study is that it lacks accuracy results, precision results and f1 scores.[18]

D. Playing cards detection

In this paper, they describe how they used YOLO V3 to detect objects with a mobile soccer robot. YOLO V3 is distinct from other object detection methods based on regional detection, such as the R-CNN series: Fast R-CNN[19] and Faster R-CNN[20]. Fast R-CNN and Faster R-CNN are two types of R-CNN. It contains a variety of potential regional generating parts and feature layers based on the region target detection approach. As a result, picture processing takes a long time, making it unsuitable for real-time applications. They choose The YOLO technique because it combines training and detection, feature extraction, and regression classification into a single network, which will be faster[21] and more useful for their purpose.[22]

A system using YOLO V3 to recognise an object at the Barelang mobile soccer robot to replace the prior method of object recognition, which was colour filtering was needed. The hardware for the Barelang mobile robot soccer team is made up of a Digital Omni Camera, a Mini Computer, a Motor DC Controller, a Motor DC, and a Kicker Selenoid Controller. A 180-degree ELP camera with a fisheye lens. They employed the darknet's pre-trained model, darnet53.conv74, which is a pre-trained model that has been trained. Imagenet single crop top-5 accuracy is 93.8 %. The proposed method gave them the IOU 71.76%, recall 0.92, precision 0.92 and mAP 87.07%[22].

The system here seeks to give an efficient technique for detecting fish and their trajectories, reducing expenditures spent by fish farmers on manual activities and providing a solution to time/labour-intensive tasks such as manually evaluating fish trajectories, allowing them to focus on their fish output. To achieve optimum detection accuracy, they opted to integrate the colour Multi-Scale Retinex colour enhancement approach with the YOLO algorithm. The YOLO object detection method detects fish, allowing them to calculate the size and number of fish. Then, to get fish trajectories, they integrated YOLO with the optical flow method. This paper is very useful as they explain the workflow in a simple clear way, but the major problem was the lack of numerical results[23].

III. Methodology

In this section, we explain in detail the workflow of our proposed system as depicted in Figure 1. Firstly, we prepare the data to be fed to the learning models in the manner that best suits each model. Then the training stage, which involves processing data and training the models. After that, fine tuning the hyper-parameters of the models. Finally, making predictions and model evaluation. In the last subsection, we introduce the data sets we used.

A. Input data preparation

- We split the videos into frames to be used in the fighting detection model.
- For Yolov5 model, we performed object of interest manual segmentation and then labeling the objects of interest to be used in the playing cards and smoking detection models.

B. Models training

- *Processing:* The system then processes the data after finishing the preparation phase. Detecting Playing-Cards and cigarette is the process that is done by using the YOLO algorithm. Then fighting is detected by long short term memory algorithm.
- *Training:* In training our YOLO model, it took 50 epochs to reach the end of training since the average loss percentage was nearly the same for 30 epochs. YOLOv5m weights were used to train the dataset (images and the text files) over the YOLO weights since it was designed for the usage of only one class which is our case and it is a smaller and lighter version of YOLOv5L which is designed for a larger number of classes. Regarding training the LSTM model,it took 25 epochs to reach the end of training by configuring its hyper-parameter with 100 batch size, and 512 RNN size.

C. Detection

YOLO algorithm which we used to detect our customised objects is a regression-based algorithm. In one run for the algorithm, YOLO calculates the region of interests and classes for the image. The algorithm gives confidentiality back when detecting an object, Firstly, one neural network is processed

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 04,2026 at 03:57:59 UTC from IEEE Xplore. Restrictions apply.

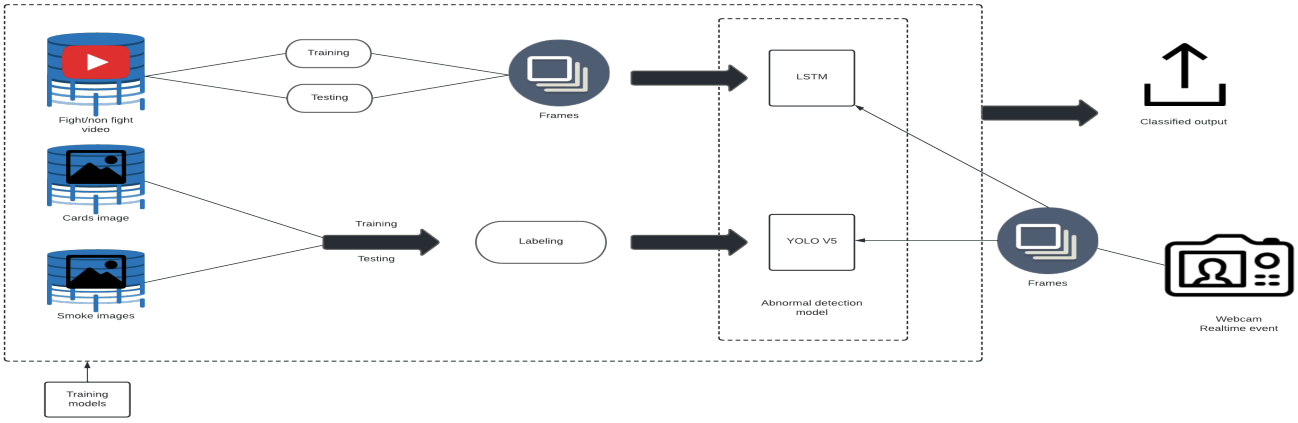


Fig. 1. The workflow of the proposed system

on the entire image. The image is then divided into different cells and the objects are predicted in each cell. The algorithm surpasses peers (e.g: RCNN, Faster RCNN) as it configures a global image context view, for prediction of the object in a single image, it takes YOLO one neural network while RCNN consumes a large number of neural networks. YOLO object detection does not hold and treat objects as the same in each video frame. So, each card and cigarette detected across video frames is assigned a specific label/ID. takes each detected card and cigarette centre points in each frame which is delivered to an array, where the coordinates of the last 8 frames are stored. Then, distance is measured (euclidean distance) between the card and cigarette current coordinates and previous frames coordinates.

The algorithm LSTM is based on an artificial recurrent neural network (RNN), that calculates with feedback connection which has a cell as an input gate, and that cell remembers values across arbitrary time intervals, which helps in data stream like videos in the fighting data-set. There are three gates in a typical LSTM; forget gate, input gate and output gate. As a result of the image recognition process, it extracts features from each image.

D. Datasets

- Hockey fight videos dataset: This dataset contains 1000 videos, The videos are divided into 500 videos fighting, and 500 are not fighting, the dataset is used in the code via URL linked to the code and the algorithm uses 80% of the dataset for training and 20% for testing[24][25].
- Cigarette smoker detection dataset: This dataset contains 3275 images, The images are divided into 1279 not smoking images, and 1996 smoking images, the dataset used in the code and the algorithm uses 70% of the dataset for training and 30% for testing[26][16].
- Playing cards dataset: it is a customized dataset made for playing cards detection that contains 3100 images of different playing card shapes front and back, and concatenating a part of a playing card dataset and a part of the UNO dataset with some customized images of Monopoly deal cards which are labelled manually all with the same class Playing Card, the dataset has 80% training, 10% validation and 10% testing[27].

IV. Experimental Results

All our experiments are conducted on a machine with specifications of an 8 GB RAM and Intel Core i7-6700HQ CPU 2.60 GHz laptop supported by an embedded web cam for the live capture of testing scenarios.

A. Experiment I- *Detection of Fights*

The used LSTM model in fights detection is trained over 25 epochs reaching loss of 0.06383, and achieving accuracy of 93.5% for classification. Figure 2 depicts the detection of a fight with confidence of 95% .



Fig. 2. Fighting model

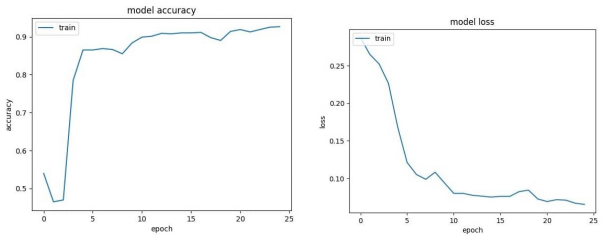


Fig. 3. Performance Evaluation of LSTM Model

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 04,2026 at 03:57:59 UTC from IEEE Xplore. Restrictions apply.

B. *Experiment II-* ***Smoking Detection***

The Yolov5 model used for detecting smoke and cigarettes was trained on customised objects such as lit and not lit cigarettes, vapes and e-cigarettes. Figure 4 depicts a cigarette detection with confidence of 73%. The snippet was taken from a live scene detection.



Fig. 4. Cigarette Detection using Yolov5 Model

C. *Experiment III-* ***Detection of Playing Cards***

Also for detecting playing card games, we used YOLOv5 model which was trained on customised objects such as standard deck of cards and UNO cards. Figure 5 depicts the detection of a cards with a confidence of 84-88%. The snippet is taken from in a live scene. It should be noted that the model is capable of detecting cards from behind even if it appears partially.

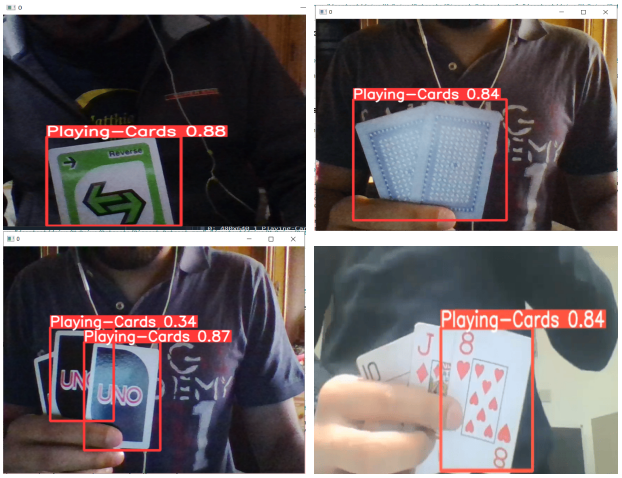


Fig. 5. Playing-cards model

The performance of Yolov5 for detecting our customised objects is depicted in Figure 6. The figure shows the loss function of training and validation sets, the means average precision over different IoU thresholds from 0.5 to 0.95 and the precision and recall of the training set, and key metrics versus confidence of the testing set.

V. Conclusion

In the last few years, researchers' attention has been drawn to the detection of violent and unwanted actions, and as a result, numerous surveillance systems have been developed. In this paper, we proposed a learning-based surveillance system consisting of three modules which all are based on deep learning to detect abnormal behaviours. The smoking module, playing cards module, and the violence module used different learning models such as LSTM and YOLO achieving accuracy above 93%. In future work, we are looking forward to fusing all the different used datasets into a single dataset to be fed to a consolidated learning algorithm. Also, we aim to enhance the accuracy of the detection of the playing cards and smoking classes. We will work on extending the system with a facial recognition module in order to be able to recognise the rule-breaker students. Finally, we need to conduct real environment testing on Campus with different test case scenarios.

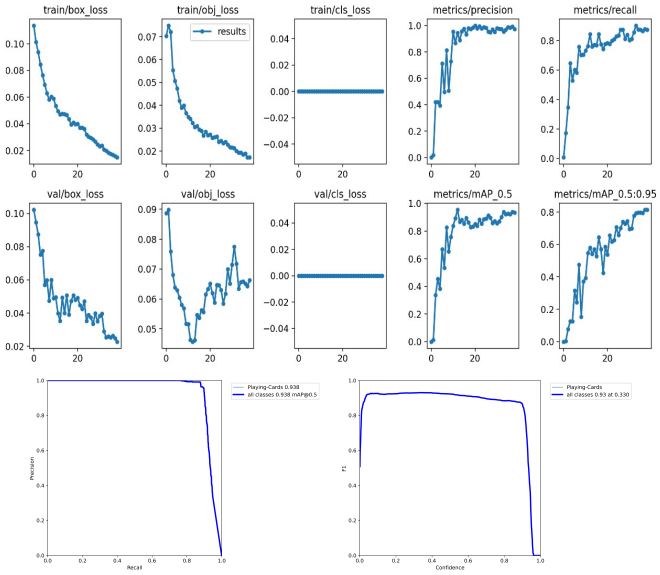


Fig. 6. Yolov5 Performance Evaluation

References

[1] Thomas Gatt, Dylan Seychell, and Alexiei Dingli. "Detecting human abnormal behaviour through a video generated model". In: *2019 11th International Symposium on Image and Signal Processing and Analysis (ISPA)*. 2019, pp. 264–270. DOI: 10.1109/ISPA.2019.8868795.

[2] Kirti Wankhede, Bharati Wukkadada, and Vidhya Nadar. "Just Walk-Out Technology and its Challenges: A Case of Amazon Go". In: *2018 International Conference on Inventive Research in Computing Applications (ICIRCA)*. 2018, pp. 254–257. DOI: 10.1109/ICIRCA.2018.8597403.

[3] Renshu Wang, Jing Zhao, Wenbin Wu, et al. "Recognition and Locating of Damaged Poles in Distribution Network Through Images Shot by Unmanned Aerial Vehicle (UA V)". In: *2020 IEEE International Conference on Information Technology, Big Data and Artificial Intelligence (ICIBA)*. Vol. 1. IEEE. 2020, pp. 1048–1052.

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 04,2026 at 03:57:59 UTC from IEEE Xplore. Restrictions apply.

[4] Joseph Redmon, Santosh Divvala, Ross Girshick, et al. "You only look once: Unified, real-time object detection". In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016, pp. 779–788.

[5] Martin Sundermeyer, Ralf Schlüter, and Hermann Ney. "LSTM neural networks for language modeling". In: *Thirteenth annual conference of the international speech communication association*. 2012.

[6] Felix A Gers, Nicol N Schraudolph, and Jürgen Schmidhuber. "Learning precise timing with LSTM recurrent networks". In: *Journal of machine learning research* 3.Aug (2002), pp. 115–143.

[7] Kazi Atai Kibria, Asswad Sarker Noman, Md. Abir Hossain, et al. "Creation of a Cost-Efficient and Effective Personal Assistant Robot using Arduino amp; Machine Learning Algorithm". In: *2020 IEEE Region 10 Symposium (TENSYMP)*. 2020, pp. 477–482. DOI: 10.1109/TENSYMP50017.2020.9230773.

[8] Kushal M, Kushal V, Charan J, et al. "ID Card Detection with Facial Recognition using Tensorflow and OpenCV". In: July 2020, pp. 742–746. DOI: 10.1109/ICIRCA48905.2020.9183342.

[9] Sandeep Kumar Srivastava, Sandhya Katiyar, and Sanjay Kumar. "Pattern Matching Using Face Recognition System". In: *Soft Computing: Theories and Applications*. Springer, 2022, pp. 161–171.

[10] S. Blunsden and R.B. Fisher. "The BEHAVE video dataset: Ground truthed video for multi-person behavior classification". In: 2010.4 (2010), pp. 1–12.

[11] Robert Fisher, Jose Santos-Victor, and James Crowley. "CAVIAR test case scenarios". In: *URL http://homepages. inf. ed. ac. uk/rbf/CAVIAR* (2004).

[12] Ersin Esen, Mehmet Ali Arabaci, and Medeni Soysal. "Fight detection in surveillance videos". In: *2013 11th International Workshop on Content-Based Multimedia Indexing (CBMI)*. 2013, pp. 131–135. DOI: 10.1109/CBMI.2013.6576569.

[13] Sarah Almeida Carneiro, Gabriel Pellegrino da Silva, Silvio Jamil F. Guimaraes, et al. "Fight Detection in Video Sequences Based on Multi-Stream Convolutional Neural Networks". In: *2019 32nd SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI)*. 2019, pp. 8–15. DOI: 10.1109/SIBGRAPI.2019.00010.

[14] Enrique Bermejo Nievas and Suarez. *Movies fight detection dataset*. URL: https://academictorrents.com/details/70e0794e2292fc051a13f05ea6f5b6c16f3d3635.

[15] Aqib Mumtaz, Allah Bux Sargano, and Zulfiqar Habib. "Violence Detection in Surveillance Videos with Deep Network Using Transfer Learning". In: *2018 2nd European Conference on Electrical Engineering and Computer Science (EECS)*. 2018, pp. 558–563. DOI: 10.1109/EECS.2018.00109.

[16] Dongyan Zhang, Cheng Jiao, and Shuo Wang. "Smoking Image Detection Based on Convolutional Neural Networks". In: *2018 IEEE 4th International Conference on Computer and Communications (ICCC)*. 2018, pp. 1509–1515. DOI: 10.1109/CompComm.2018.8781009.

[17] Junaidy B. Sanger, Lanny Sitanayah, and Vivie D. Kumenap. "Detection System for Cigarette Smoke". In: *2019 4th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE)*. 2019, pp. 145–149. DOI: 10.1109/ICITISEE48480.2019.9003901.

[18] Yue Chang, Zecheng Du, and Jie Sun. "Dangerous Behaviors Detection Based on Deep Learning". In: *Proceedings of the 2nd International Conference on Artificial Intelligence and Pattern Recognition*. AIPR '19. Beijing, China: Association for Computing Machinery, 2019, pp. 24–27. ISBN: 9781450372299. DOI: 10.1145/3357254.3357267. URL: https://doi.org/10.1145/3357254.3357267.

[19] Ross Girshick. "Fast R-CNN". In: *Proceedings of the 2015 IEEE International Conference on Computer Vision (ICCV)*. ICCV '15. USA: IEEE Computer Society, 2015, pp. 1440–1448. ISBN: 9781467383912. DOI: 10.1109/ICCV.2015.169. URL: https://doi.org/10.1109/ICCV.2015.169.

[20] Shaoqing Ren, Kaiming He, Ross Girshick, et al. "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks". In: *Advances in Neural Information Processing Systems*. Ed. by C. Cortes, N. Lawrence, D. Lee, et al. Vol. 28. Curran Associates, Inc., 2015. URL: https://proceedings.neurips.cc/paper/2015/file/14bfa6bb14875e45bba028a21ed38046-Paper.pdf.

[21] Joseph Redmon and Ali Farhadi. "YOLOv3: An Incremental Improvement". In: *ArXiv* abs/1804.02767 (2018).

[22] Hendawan Soebhakti, Senanjung Prayoga, Rifqi Amalya Fatekha, et al. "The Real-Time Object Detection System on Mobile Soccer Robot using YOLO v3". In: *2019 2nd International Conference on Applied Engineering (ICAE)*. 2019, pp. 1–6. DOI: 10.1109/ICAE47758.2019.9221734.

[23] Youssef Wageeh, Hussam El-Din Mohamed, Ali Fadl, et al. "YOLO fish detection with Euclidean tracking in fish farms". In: *Journal of Ambient Intelligence and Humanized Computing* 12.1 (2021), pp. 5–12.

[24] YASSER SHRIEF. *"Hockey Fight Vidoes"*. Feb. 2021.

[25] Oluwatoyin P. Popoola and Kejun Wang. "Video-Based Abnormal Human Behavior Recognition—A Review". In: *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* 42.6 (2012), pp. 865–878. DOI: 10.1109/TSMCC.2011.2178594.

[26] Han Lee, aus10powell, Arsen. *Cigarette Smoker Detection*. Kaggle, May 9, 2019. URL: https://www.kaggle.com/vitaminc/cigarette-smoker-detection.

[27] James McGuigan. *Playing Cards*. Kaggle, July 14, 2020. URL: https://www.kaggle.com/jamesmcguigan/playingcards.

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 04,2026 at 03:57:59 UTC from IEEE Xplore. Restrictions apply.